P4 confirmatory study (prime medium tier): results

Study research/prime_medium_rho_p4_confirm_20261108, protocol frozen 2026-11-08T05:12:02Z (protocol_sha256 26fd5c57…ae0c, commit de0e7026b0) before any data. Every number is [pcc:unverified]; sources are named in RESULTS.md and p4conf_summary.json.

Verdict (PROTOCOL.md section 9, as written)

inconclusive: gate G3_positive_control failed. The positive control bsgs.textbook / bsgs.negation reads 1.4639, 95 % [1.4438, 1.4863], 99 % [1.4384, 1.4938]. Its 95 % lower end clears 1.2, but the expected 1.5 is outside the 99 % interval, and G3 requires both. Any failed gate makes the verdict inconclusive.

```
G1_sessions_complete_and_audited   pass
G2_aa_determinism                  pass
G3_positive_control                FAIL
G4_split_half_aa                   pass
G5_censoring                       pass
G6_pair_coverage                   pass
```

Flow: 2160/2160 pairs complete, A/A mismatches 0 (main) and 0 (PC), censored share 0 in every arm.

Primary, descriptive only (it does not decide anything here)

theta (w = 0) = 1.3624, 95 % [1.3202, 1.4054], half-width 0.0426. The primary_outcome field reads exceeds_1.2_below_sqrt2. It excludes the parent study's 1.285 and sqrt2, contains the look-ahead heuristic 1.371, and its point lies inside the parent J1 interval [1.177, 1.404]. Because G3 failed, none of this is a verdict.

Positive-control mechanism (exploratory, not part of the frozen analysis)

In every one of the 96 PC targets, S_tb - S_neg lies in [0.49996, 0.50000]: both BSGS variants take the same number of giant steps (stride about sqrt r) and differ only by their baby tables. The PC ratio is therefore 1 + 0.5 / mean S_neg, and it measures the mean planted scalar d/r of its 96 targets. That mean is 0.5776, not 0.5 (z = 2.63; KS p about 0.007). The exact uniform-d expectation of the ratio at these curve sizes is 1.49993, so set-up and finite size explain under 1e-4 of the 0.036 shortfall. The 360 main-job targets, drawn by the same law, have mean 0.4990 (KS p about 0.95). The law hashes r, seed and index (workload.rs at dbd587a01a).

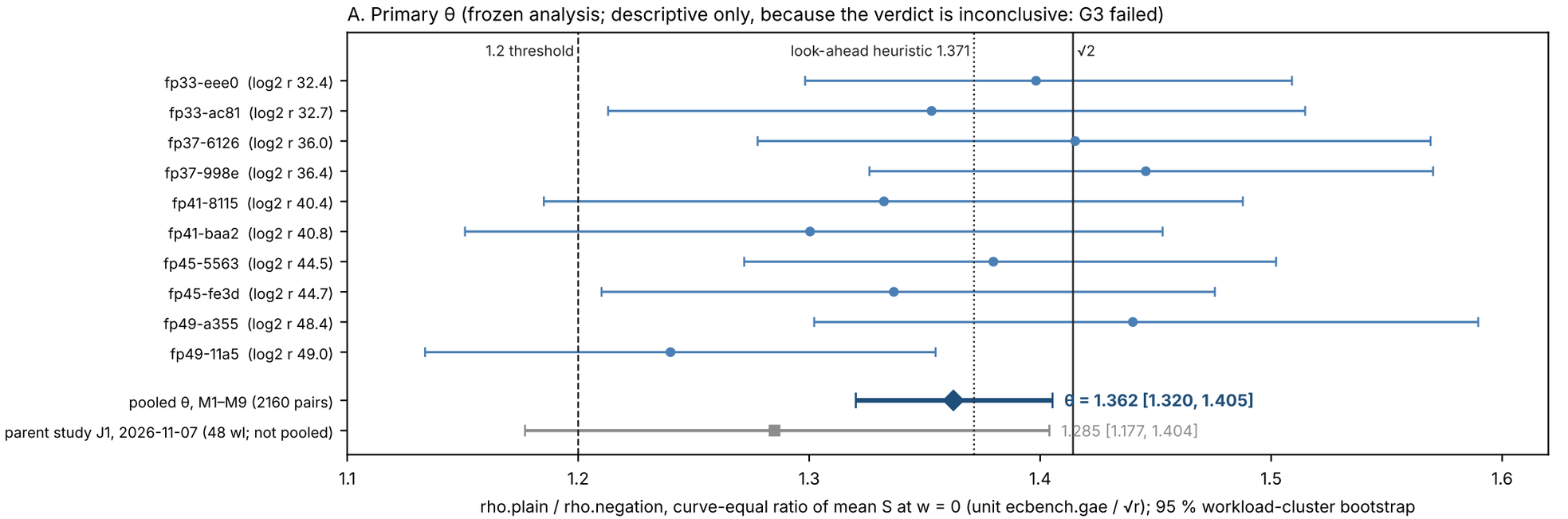
Under uniform targets, a ratio at or below 1.4639 occurs in 0.41 % of 20000 simulated designs. A design-level simulation of G3 itself (2000 designs, 4000 resamples each) fails the 99 % condition in 1.8 % (MC s.e. 0.3 %), against a nominal 1 %. Hypothesis, untested by any new run: the shortfall is a low-probability draw of the PC target block, not a harness or estimator defect.

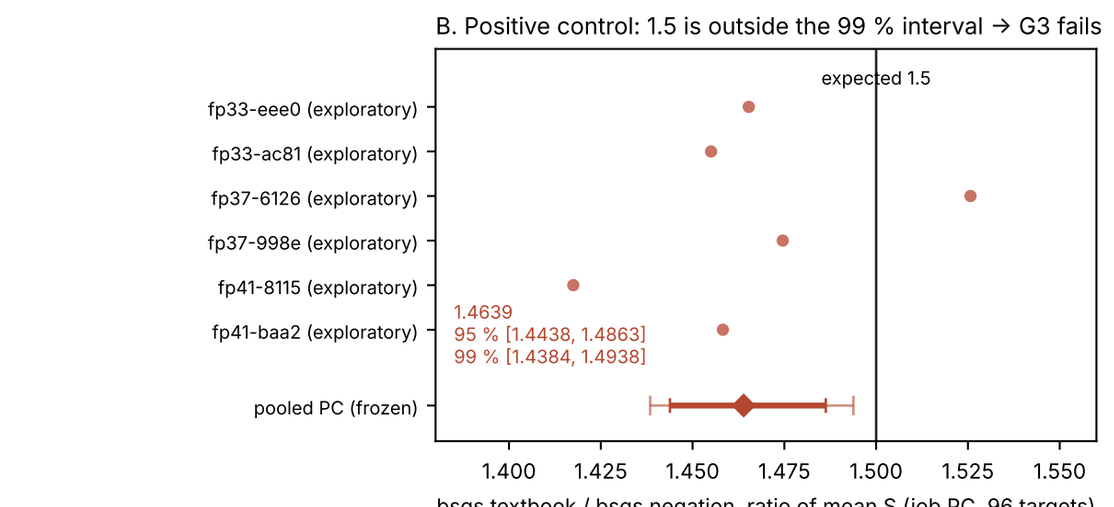
Execution

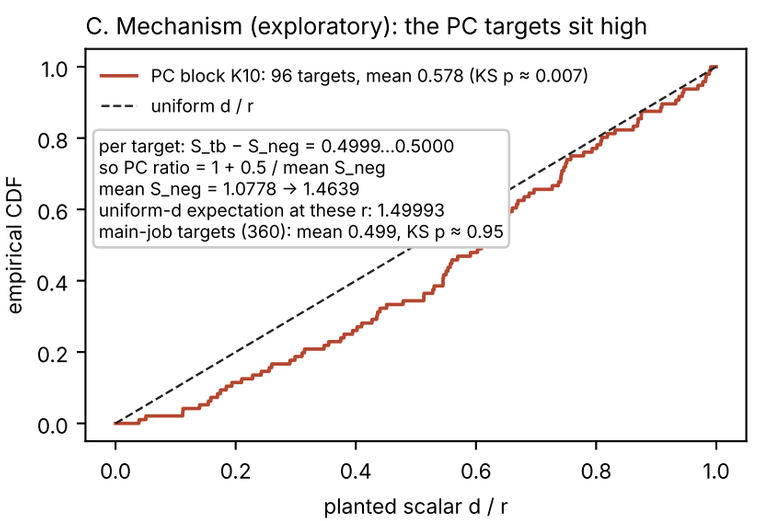
```
PC  pod exit 43  exec  2016  replays 1728  below-L2   0  audit ok True
M1  pod exit 42  exec  6048  replays 5184  below-L2  42  audit ok True
M2  pod exit 42  exec  1008  replays  864  below-L2  64  audit ok True
M3  pod exit 42  exec  1008  replays  864  below-L2  52  audit ok True
M4  pod exit 42  exec   336  replays  288  below-L2 152  audit ok True
M5  pod exit 42  exec   336  replays  288  below-L2 139  audit ok True
M6  pod exit 42  exec   336  replays  288  below-L2 145  audit ok True
M7  pod exit 42  exec   336  replays  288  below-L2 176  audit ok True
M8  pod exit 42  exec   336  replays  288  below-L2 174  audit ok True
M9  pod exit 42  exec   336  replays  288  below-L2 158  audit ok True
```

Ten jobs, once each, on c8a.2xlarge (queue cpu-l2-spot per PROTOCOL.md section 11). M1-M9 pods exited 42 (records below L2: solves under 1 ms); the PC pod exited 43 (all 2016 records L2; the later reference/control wall-comparison gate of run-ecbench did not pass, and which of its conditions failed is not in the fetched receipts). All ten are admitted for counts under section 10; no wall time is reported. Card -9 was killed during its local verify and finish card V-M9 completed it. No rerun.

P4 confirmatory study, prime medium tier: verdict inconclusive (gate G3_positive_control failed)







Sessions: PC ECBS1h920148bdc4ac; M1–M9 as in analysis/result.json. Frozen analysis analysis/p4_confirm.py (sha256 01e62709...), 20 000 resamples, seed 47393111417401. Panels B (per-curve dots) and C are exploratory (scratch scripts, scripts-archive.json). All values [pcc:unverified].

Predictions (section 12) and caveats

Predictions, written before data (graded descriptively; they are not the verdict)

P1. Predicted: theta about 1.30; interval within [1.2, sqrt2); 1.285 inside. Observed: theta 1.3624 [1.3202, 1.4054]; interval within [1.2, sqrt2): yes; 1.285 inside: False. Reading: fails: point above 1.30 and 1.285 outside the interval (interval placement as predicted).

P2. Predicted: theta_steps about 1.35, interval below sqrt2 (walk_level_shortfall). Observed: theta_steps 1.4098 [1.3660, 1.4542]; attribution charged_overhead. Reading: fails: interval contains sqrt2; attribution charged_overhead.

P3. Predicted: PC ratio 1.50 +- 0.05; lower end above 1.2. Observed: 1.4639 [1.4438, 1.4863]. Reading: holds as worded (point within 0.05 of 1.5, lower end above 1.2); gate G3 is stricter and fails.

P4. Predicted: A/A identical; split-half intervals contain 1. Observed: aa mismatches 0; split neg 0.9824 [0.9325, 1.0352], plain 1.0207 [0.9721, 1.0721]. Reading: holds.

P5. Predicted: about 1.18 at w=0.1 and about 0.65 at w=1; order unresolved. Observed: w=0.1 1.2388 [1.2003, 1.2778]; w=1 0.6817 [0.6606, 0.7032]; unresolved inside the band. Reading: order holds; both point predictions lie below the measured intervals.

P6. Predicted: size-slope interval contains 0. Observed: -0.00294 [-0.00816, 0.00216]. Reading: holds.

P7. Predicted: theta_v2 within 0.05 of theta. Observed: theta_v2 1.3876 vs theta 1.3624 (difference 0.0252). Reading: holds.

Caveats

No independent validator has re-derived these numbers, and no pcc claim exists: everything is [pcc:unverified].

The exploratory analyses use scratch scripts outside the repository (scripts.tar.gz in S3, sha256 in scripts-archive.json). The G3 calibration uses numpy's RNG and 4000 resamples, not the frozen script's resampler.

The receipts record c8a.2xlarge; the queue name cpu-l2-spot comes from PROTOCOL.md section 11 and the request routing, not from a receipt field.

research-kb and Conductor are not installed in this environment; the literature in PROTOCOL.md section 9a stays 'recalled'.